



**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani Pilani
Campus
AUGS / AGSR Division**

**FIRST SEMESTER 2025-2026
COURSE HANDOUT (Part-II-B)**

Date: 02/08/2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course

Course No. : PHA F313
Course Title : Instrumental Methods of Analysis
Instructor-in-Charge : Prof. Jagpreet Sidhu
Instructors : AS GIVEN IN TIME TABLE

No of Lecture, Tutorial Hours per week: 03 No. of Practical hours per week: Three (1 X 3)

1. Course Description:

This course is a laboratory oriented course and covers instrumentation and applications of various analytical techniques such as UV-visible, Infrared, NMR spectroscopy, Mass spectrometry etc for drug discovery and development

2. Scope and Objective of the Course:

The course is aimed at exposing the students to modern analytical techniques in relevance to pharmaceutical industries. Training with respect to the operation of sophisticated instruments, interpretation of results obtained for the characterization of molecules, their qualitative and quantitative evaluation are emphasized.

Objectives

- ⇒ Exposure to principles, instrumentation, calibration and operation of analytical instruments
- ⇒ Obtaining spectral data, their interpretation, evaluation of qualitative and quantitative aspects of drugs and their estimation

3. Text Books:

1. Williard H.H., et al., "Instrumental Methods of Analysis", CBS Pub., New Delhi, 7th ed., 1988
2. Parimoo P, -"Instrumental Methods of Analysis", CBS Pub., New Delhi, 1988

4. Reference Books:

1. Ewing, Galen Wood. - "Instrumental Methods of Chemical Analysis", 5th Edition, Mc.Graw- Hill, Inc., N.Y., 1985.
2. Robert M Silverstein, Francis X Webster -"Spectroscopic Identification of Organic Compounds", Sixth Edition, John Wiley and Sons, Inc., N.Y., 2002.
3. A.H.Beckett and J.B. Stenlake -"Practical Pharmaceutical Analysis", 3rd Edition, CBS Pub., NewDelhi., Vol. I & II, 1968
4. Skoog. Holler. Nieman. "Principles of Instrumental Analysis", 5th Edition, Thomson Publishers, NewDelhi, 2006.
5. B. K. Sharma. "Instrumental Methods of Chemical Analysis", 27th Edition. Goel Publishing House, Meerut. 2011.
6. Donald L. Pavia, Gary M. Lampman, George S. Kriz, James A. Vyvyan. "Introduction to Spectroscopy", 4th Edition, Brooks Cole Publishers.
7. R. Mahesh, Sajeev C, N. Sridhar, Laboratory manual on "Instrumental Methods of Analysis" – EDD Notes. 4th Edition., 2003.

5. Course Plan:

Lec. No	Learning Objectives	Topics to be covered	References	Learning Outcome
1-4	Introduction to Instrumental Methods of Analysis	Terms associated with chemical analysis Classification of	TB-1: 1-5; TB-2: 1, 2	Student will understand basics of analytical



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		instrumental Techniques Basic functions of Instruments		techniques and instrumentation concepts
5-8	UV-Visible Spectroscopy	Basic principle, Radiation source, Wavelength selection, Cells and sampling devices, Detectors, Fundamental laws of photometry, Woodward Fischer Rule, Prediction of λ_{max} of molecules	TB-1: 6, 7; TB-2: 14, 15	Students will learn the application of UV spectroscopy for structure elucidation and quantitative estimations of drug molecules
9-13	IR Spectroscopy	Introduction, Theory, Instrumentation, Sample handling, Structural analysis of organic compounds	TB-1: 11; TB-2: 18	Students will get acquainted with IR methods for functional group identification and thus their structure elucidation
14-16	Spectro fluorimetry	Introduction, Structural factors, photoluminescence power as related to concentration, Instrumentation, Fluorescence measurement	TB-1: 8; TB-2: 17	Students will understand the fluorescence phenomenon and its application for fluorescent drugs/ligands
17-18	Polarimetry	Introduction, theory, calculation, applications	TB-2: 10	Students will understand the conversions of isomers
19-22	NMR Spectroscopy	Introduction, theory, Instrumentation, structure elucidation of organic molecules, applications	TB-1: 15; TB-2: 19	Students will be able to use the nuclear spin property of elements for structure elucidation
23-26	Mass Spectroscopy	Basic principle, Instrumentation, Fragmentation of organic	TB-1: 16; TB-2: 20	Students will learn application of high energy sources such as electron, atoms etc for structure elucidation of organic molecules including drugs



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27-28	Flame photometry & Atomic absorption spectrometry	Theory, instrumentation methodology, sources, applications	TB-1: 9, 10; TB-2: 16	Students will learn to use flame as a source for estimation of metallic elements in samples
29-38	Chromatographic techniques – Introduction	Terminology used, principles, Classification, Ion-exchange, Size exclusion, Paper, Gas Chromatography, High Performance Liquid Chromatography, TLC, High Performance Thin Layer Chromatography (HPTLC), High Voltage paper / Gel Electrophoresis, Basics of Hyphenated Techniques	TB-1: 17; TB-2: 21 TB-1: 18; TB-2: 21 TB-1: 19, 20; TB-2: 21 TB-2: 21, 22 RB-4-20, RB-3	Students will understand the various techniques for separation, purification of drug molecules and their quantitative estimation
39-41	Thermography	General aspects, components and methodology, uses of DSC, TGA, DTA	TB-1; TB-2	Students will learn to use and interpret DSC, TGA, DTA for identification of effect of heat of physical and chemical transformation of drugs

Practicals

S. No. Name of Experiment

- UV - Visible**
 - Determination of λ_{max} , verification of Beer-Lambert's Law
 - Determination of binary mixtures
 - Difference, derivative spectroscopy
 - Unknowns - interpretation and analysis
- IR**
 - Handling solid, liquid, gaseous samples
 - Effect of solvents on IR absorptions
 - Unknowns - interpretation, functional group analysis
- Polarimetry**
 - Determination of muta-rotation in glucose, unknown
 - Estimation of specific rotations for a few selected drugs
 - Evaluation/analysis of isomers in mixtures
- Spectro-fluorimetry**
 - Estimation of selected drugs compounds, unknown
 - Quenching effect and its significance
 - Evaluation of drugs in formulations, interpretation
- Flame photometry and A.A.S***
 - Calibration, estimation of elements, applications
- Chromatography (HPLC)**
 - Instrumentation, method development approaches
 - Separation and estimation of selected drug samples
 - Unknown identification, applications



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Chromatography*

(GC)

- d) Instrumentation, method development approaches
- e) Separation and estimation of selected drug samples
- f) Unknown identification, applications

7. **Thermography***

- a) Instrumentation, calibration of DSC / TGA
- b) Unknown-identification, applications

(*These experiments would be demonstrated / performed depending on time availability either as a regular experiment or as a part of an assignment) Note:

1. Modifications / adjustments would be made in the theory/experimental pattern / part, if necessary as and when situation arises.
2. Students should maintain and bring updated record note-books for every practical class.
3. Make-ups for practical are not always possible. However, depending upon the genuineness of the situations, students may be permitted to perform back-log experiments, if any instructor is free, outside regular class hours.
4. It is imperative that all students come prepared for the experiment in the next turn completing all pending work concerned with the previous experiment. Adequate preparation for the practical in terms of principles and operation of the instrument as per instructions, familiarization of protocols involved, outside class hours, is mandatory.

6. **Evaluation:**

S. No.	Evaluation Component	Duration	Weightage (%)	Date & Time	Remarks
1.	Mid-sem Exam	90 min	30		CB
2.	Continuous assessment		35		CB & OB (or) CB / OB
3.	Comprehensive Exam	180 min	35		CB & OB (or) CB / OB

Note: Continuous assessment will be based on theory covered in class. Topics and number will be announced in the class. It will be based on home assignments, tutorials, quiz, seminar, laboratory, viva- voce and class participation.

7. **Chamber consultation hours:** To be announced in class.

8. **Notices:** Notices concerning the course will be displayed on the Department of Pharmacy N.B. only.

9. **Make-Up Policy:** Generally, make-up will be considered for regular students only and Make-Ups are not given as a routine. It is solely dependent upon the GENUINENESS OF THE CIRCUMSTANCES under which a student fails to appear in a scheduled evaluation component. In such circumstances, prior permission should be obtained from the Instructor-in-Charge. IN NO CASE THE MAKE-UP APPLICATION BE SLIPPED INSIDE THE CHAMBER OF THE INSTRUCTOR-IN-CHARGE. However, the decision of the Instructor-in-Charge in the above matter will be final.

Instructor-in-Charge
PHA F313



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